

AIBS - Handbook

Group 7

Target audience

- SME's handbook is for sports medium sized Dutch company like Deventrade.
- It is a family owned company.
- They make teamwear and gear for football, hockey, handball, and volleyball, sold under a handful of brands they own, like Hummel, Stanno, Reece, Derbystar, and Select.
- They could increase their market share through technological development therefore we write the handbook to anyone who could profit from it.

What they need

- Companies like this should make the decision about where they can implement AI relatively cheap.
- The stakeholders should learn about the best use of AI in their field.
- These companies should always make sure that they are using any assistance that their government can give.
- They need clear, realistic criteria to distinguish high-impact, practical AI tools from expensive industry hype before committing limited company capital.

Research question

- Main research question:

We wanted to know whether AI-based demand forecasting has actually gotten mature enough for mid-sized, seasonal wholesalers with a lot of product variants to realistically use it alongside the planning tools they already run, or if it's still more promise than something you could actually roll out in the next year or two.

- Sub-questions:

Market/external:

- *Who is actually building AI demand-forecasting tools for companies this size right now, and are they building for SMEs or mostly chasing the big retail chains?*
- *What do companies that already tried this say went wrong? What are the real failure points, not just what the vendors put in their pitch decks?*

Internal:

- *How good does a company's historical sales data typically need to be before a model can actually learn anything useful from it?*
- *Would a company like this need to hire someone new to own this long-term, or could it realistically run with the team they already have?*

Quality criteria

- Help readers decide whether to use an existing AI tool, customise current software or develop a new solution. Existing tools should generally be tested before committing to custom development.
- Tailored specifically to the realities of pre-season club cycles, batch manufacturing lead times, and broken size curves rather than generic e-commerce advice.
- Openly details common failure points (e.g., bad historical ERP inputs, black-box distrust from sales staff, rigid vendor contracts) rather than marketing claims.
- Tailored specifically for a business with 100–200 staff and a 1–2 person IT department. Every recommendation must specify transparent price bands (setup cost, monthly SaaS fees, internal hours required) and evaluate off-the-shelf software before suggesting custom development.
- Describe what companies should do after a pilot succeeds, including improving the tool, connecting it to existing systems, training more users and monitoring performance.
- Identify who is responsible for the project, the data, technical implementation, risk management, employee training and final decisions.
- Address privacy, security, bias, transparency, human oversight and accountability. The handbook should make clear which decisions require human review.

Methodology

- **Desk research**
 - We are mostly working from what is already out there for this part, vendor sites, product docs, a handful of case studies from companies that tried something similar. That's where we will find out who is actually building AI forecasting tools for companies in our market, versus who is just chasing the big retail chains. It is also the part that tells us what actually goes wrong when people try this in practice, not just what the sales pitch says. And honestly, some of the "how much data do you even need" question can be answered this way too, at least in general terms. There is enough written about typical data requirements that we don't need to figure that out from scratch.

- **Field research (site visit)**
 - This is where the real answers are, at least for the two questions that are actually about them specifically. No amount of reading is going to tell us what their sales data actually looks like day to day, or whether they have got someone who could realistically run something like this once it is built, versus needing outside help forever. That's a conversation, not a search.

- Practically, it makes sense to get the desk research mostly out of the way before the visit happens. That way, when we are actually there, we are not wasting time on stuff we could have googled, and we can spend it asking the questions only they can answer.